

Nihal Singh

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PROFESSIONAL SUMMARY

Computer Science undergraduate at BINUS University building applied machine learning projects across computer vision and NLP, including gesture recognition, facial emotion classification, and transformer-based text summarization. Looking for an internship to build and grow under an experienced engineering team.

EDUCATION

Binus University

Bachelor of Computer Science -Intelligent Systems

September 2024 - 2028

Jakarta, Indonesia

- Related Coursework : Algorithm & Programming, Data Structures, Natural Language Processing, Computer Vision, Machine Learning, Software Engineering, Database Technology
- GPA: 3.41/4.00

PROJECTS

American Sign Language Detection | Python, openCV

May 2026

<https://computer-vision-project-five.vercel.app/>

- Built a real-time American Sign Language recognition system using MediaPipe Hand Landmarker (21 keypoints, 63 coordinate features) and a Support Vector Classifier (RBF kernel), achieving 99.74% test accuracy after GridSearchCV tuning.
- Designed a landmark-based pipeline instead of raw-frame CNN processing, cutting compute requirements while keeping inference real-time on consumer webcam hardware.
- Applied z-score normalization and rotational augmentation to landmark coordinates to reduce sensitivity to hand position, scale, and orientation.

Lightweight Facial Emotion Recognition with EfficientNet-B0 (FER2013) | Python, openCV

May 2026

<https://drive.google.com/file/d/1qaEBek12Dp-6nTt8aLEQ8M4bB1s7oCj5/view?usp=sharing>

- Designed a computationally efficient facial emotion recognition model combining EfficientNet-B0, squeeze-and-excitation attention, and a facial landmark branch, achieving 71.45% test accuracy on FER2013 (7 classes, 7,178 test images) with only 6.03M parameters.
- Outperformed a self-built ResNet50+SE baseline (69.94% accuracy, 102.55M parameters) while using 17x fewer parameters, and matched 97% of a VGGNet reference paper's accuracy (73.28%) at roughly a quarter of its parameter count.
- Applied CutMix augmentation and test-time augmentation to improve generalization and prediction stability under FER2013's class imbalance.
- Identified through confusion matrix analysis that negative emotions (Fear, Sad, Angry) remain the primary error source due to low resolution and overlapping facial expressions, informing future work on class balancing and landmark refinement.

Indonesian News Summarization with Transformer Models | Python, Torch, NLTK

June 2026

<https://nikiloves-indonesian-news-summarization.hf.space/>

- Prepared and validated the Indonesian XL-Sum dataset for abstractive summarization, checking label consistency between source articles and reference summaries before fine-tuning.
- Fine-tuned and benchmarked four transformer architectures (T5, BART, mT5, PEGASUS) against the validated dataset, comparing outputs against ground-truth summaries to assess model reliability.

- Built and deployed an interactive Streamlit app on Hugging Face Spaces letting users select a model and generate summaries from raw article text, exposing model behavior for direct evaluation.

Academic Performance Prediction from Sleep Patterns | Scikit-learn, Seaborn, Pandas

June 2026

<https://machinelearningsleepz.streamlit.app/>

- Executed exploratory data analysis across a dataset of 634 students using correlation matrices, distribution profiling, missing-value checks, and anomaly detection to evaluate sleep habits against academic performance.
- Built and evaluated a Gradient Boosting Regressor using 5-fold cross-validation, achieving 0.30 MAE and 0.256 R^2 on GPA prediction.
- Analyzed feature importance, identifying prior GPA (51.8%) and combined sleep-quality features (40.4%) as the strongest predictive signals, then deployed the model as an interactive Streamlit application.

EXPERIENCE

Web Development Member

2026 – present

Himpunan Mahasiswa Teknik Informatika

Jakarta, Indonesia

- Participated in daily standups and sprint discussions with a 5+ member web development team, tracking task progress and surfacing blockers.
- Reviewed peer code and provided structured feedback on HTML/CSS/JavaScript implementations before merge, improving consistency across team contributions.
- Documented setup steps and coding conventions for onboarding new members, reducing repeated troubleshooting questions in the team channel.
- Contribute to team discussions and problem-solving activities during project development

Chairperson, Church Inauguration Committee

2026

Gereja Pantekosta di Indonesia (GPdI)

Kabupaten Bogor, Indonesia

- Led a committee of more than 25 volunteers in planning and executing a church inauguration event.
- Coordinated team members across different age groups, ensuring effective communication, task delegation, and collaboration throughout the preparation process.
- Managed relationships and communications with distinguished guests, including church leadership and regional religious representatives.
- Successfully delivered the event according to plan, contributing to a smooth and well-attended inauguration ceremony

TECHNICAL SKILLS

Languages | Python, JavaScript, C, SQL, HTML, CSS

Frameworks | React, Django, Flask, Express.js

Libraries | Pandas, NumPy, Matplotlib, Torch, TensorFlow, scikit-learn, seaborn, Bootstrap